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Complex Engineering Problem

Surface Defect Detection and Classification from Images

Classifying paper-reel images as defect-free, hole or pen mark, and localising the defect with a bounding box

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Source code: `https://github.com/hbnasir23/ml-cep-ct317`

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Abstract

This project detects defects on a reel of paper from camera images. Each image must be classified into one of three classes (defect-free paper, paper with a hole, paper with a pen mark), and in defective images the defect must be localised with a bounding box. The dataset contains 1,500 images of 2000×2000 pixels.

The defects are tiny (a hole covers less than 0.01% of the image), while uneven illumination and paper texture dominate the raw pixels. We therefore designed a classical pipeline in four steps:

1. *illumination flattening*, which converts every pixel into a robust z-score;
2. *candidate proposals*, the dark regions found by thresholding and morphology;
3. *features* from the handout (colour histogram, spatial binning, HOG) and from the literature (LBP texture, GLCM statistics, blob shape descriptors and Hu moments), computed for the whole image and for the candidate region;
4. *data augmentation* by rotations and flips.

Four classifiers (SVM, Naive Bayes, Logistic Regression and a Neural Network) were tuned on a validation set and evaluated once on 225 held-out test images. All of them exceed 94% accuracy, and the best reaches **99.1%**. For localisation, a region classifier trained with hard negatives scores every candidate. The final system, selected on validation data, reaches **97.3%** test accuracy with no false alarms on clean paper. Its bounding box lies on the true defect in every inspected test image. The models are delivered in an interactive web application with single-image and batch inspection and a view of how the model sees each image. The code is public at `https://github.com/hbnasir23/ml-cep-ct317`.

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Chapter 1

Introduction

1.1 Motivation

Automatic visual inspection is essential in modern manufacturing, where defects must be found in real time at production speed. Machine-vision systems capture images of a material's surface and detect anomalies such as scratches, holes and marks. This is faster, cheaper and more consistent than manual inspection, and the collected data can be used to optimise the production process. Deep networks are common today, but a solid understanding of the classical approach (isolating an object and describing it with engineered features) is still important, and is the focus of this project.

1.2 Problem statement

Following the CEP handout, we must:

1. extract features of the object from the dataset, using the suggested features (colour histogram, spatial binning, histogram of oriented gradients) and further features identified in the literature;
2. optionally use data augmentation to obtain more training data;
3. evaluate Support Vector Machines, Naive Bayes, Logistic Regression and a Neural Network on two tasks: (i) *three-class classification* into paper, paper with a hole and paper with a pen mark; (ii) *defect localisation*, i.e. finding the defect in defective images and drawing a bounding box around it.

1.3 Contributions

- An analysis of the dataset that explains why raw global features are insufficient (tiny defects, strong lighting gradient, static lens blemishes).
- An illumination-flattening and candidate-proposal stage that makes the defects visible and isolates them as objects.
- A feature set combining the required features with LBP, GLCM, shape descriptors and Hu moments, at image level and at candidate level.
- A comparison of four classifiers on three feature sets, all above 94% test accuracy and up to 99.1%.
- A hard-negative region classifier that performs both classification and localisation, with an error analysis.

- An interactive application for single-image and batch inspection with explanations.

1.4 Tools

Python 3.13, scikit-learn 1.8 [1], OpenCV 5.0, scikit-image 0.26 [2], NumPy, pandas, matplotlib and Streamlit. Everything runs on a CPU.

Chapter 2

Dataset Analysis

The dataset, downloaded from the Google Drive folder linked in the handout, contains **1,500** JPEG images of 2000×2000 pixels: 509 *paper* (defect-free), 489 *hole* and 502 *penMark*. Inspecting the data revealed several properties that determined the design:

- The three colour channels are identical, so the images are effectively **grayscale**, and the "colour histogram" is an intensity histogram.
- The defects are **tiny**. A hole is a dark speck of about 5–25 px (less than 0.01% of the image), and a pen mark is a thin dark stroke of about 15–120 px. The paper texture and a strong **illumination gradient** dominate the image, so global features of the raw image mostly describe lighting and texture.
- A few faint dark blemishes appear at the **same position in every image** (dust on the lens or background). A detector must learn to ignore them.
- The image numbers 0–1499 are shared by all classes and interleaved randomly, so the images are not consecutive video frames, and a stratified random split is appropriate.

The images were split with stratification into **1,050 training, 225 validation and 225 test** images (70/15/15). Validation images are used for every design and hyper-parameter decision, and the test images only for the final evaluation.

Table 2.1: Dataset split.

Split	Paper	Hole	Pen mark	Total
Train	356	342	352	1,050
Validation	76	74	75	225
Test	77	73	75	225

Chapter 3

Pre-processing and Candidate Regions

1. **Down-sampling** to 1000×1000 with area interpolation, which preserves small dark spots.
2. **Illumination flattening.** The background is estimated with a wide Gaussian blur ($\sigma = 20\,\text{px}$) and subtracted: $r = I - G_\sigma * I$.
3. **Robust z-score map** $z = (r - \text{median}(r)) \,/\, (1.4826\,\text{MAD}(r))$. Clean paper texture stays within about $[-7, +10]$, while holes and ink reach -10 to -50, independently of the lighting.
4. **Candidate proposals.** Pixels with $z < -4$ (after light smoothing) are kept, a 7×7 morphological closing joins the broken strokes of a pen mark, and connected components are ranked by their "darkness mass" $\sum(-z)$.

A calibration over all 1,500 images confirmed that the classes separate: the darkest response on clean paper is $-5.9 \pm 0.6\sigma$, compared with $-30.7 \pm 10.5\sigma$ for holes and $-23.8 \pm 4.3\sigma$ for pen marks. Pen-mark regions are about 4–5 times larger than hole regions.

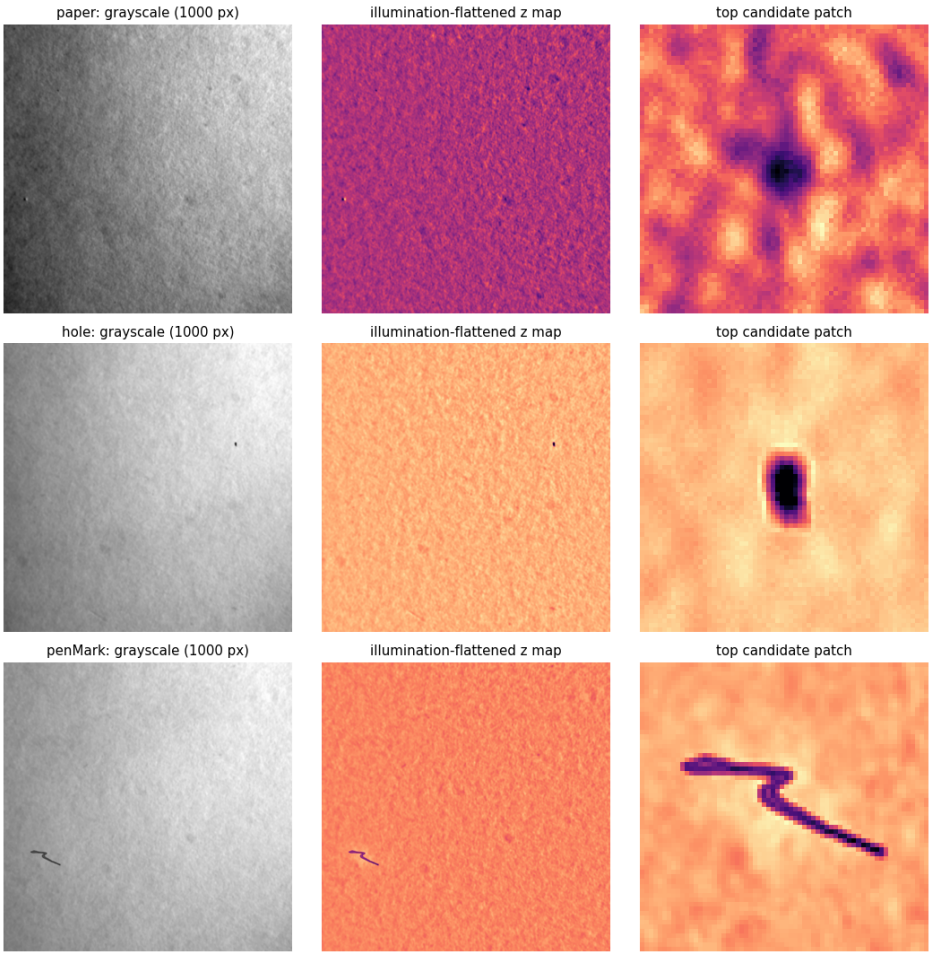


Figure 3.1: Pre-processing for a test image of each class: grayscale input, illumination-flattened z-map, and the patch around the top candidate.

Chapter 4

Feature Extraction

4.1 Literature review

Classical surface-inspection systems describe defects by intensity statistics, texture and shape [3, 4]. In addition to the handout's colour histogram, spatial binning [5] and HOG [6], we used:

- **Local Binary Patterns (LBP)** [7], a grey-scale-invariant texture descriptor widely used for surface and fabric defect detection;
- **Grey-Level Co-occurrence Matrix (GLCM)** statistics [8] (contrast, dissimilarity, homogeneity, energy, correlation), the classical texture features for steel, wood and paper inspection;
- **shape descriptors** of the segmented blob (area, aspect ratio, extent, solidity, eccentricity, major and minor axis, circularity) and **Hu moment invariants** [9], which directly encode "compact spot" versus "elongated stroke".

4.2 Implemented features

Table 4.1: Feature groups (4,959 features in total).

Set	Group	Dim.	Description
Global	Colour histogram	32	intensity histogram of the image
	z histogram	32	log-count histogram of the z-map
	Spatial binning	1,024	z-map min-pooled to 32×32
	HOG	1,764	HOG of the z-map (128×128, 16 px cells, 2×2 blocks, 9 bins)
Patch	z histogram	16	histogram of the candidate patch
	Spatial binning	256	patch resized to 16×16
	HOG	1,764	64×64 patch, 8 px cells, 2×2 blocks, 9 bins
	LBP	10	uniform LBP ($P = 8$, $R = 1$) histogram
	GLCM	40	5 properties × 2 distances × 4 angles
	Shape	21	blob geometry, darkness statistics, 7 Hu moments, patch scale

The patch is a square crop centred on the candidate, with a side of $\max(64, 1.4\times$ the blob size), resized to 64×64. This keeps the whole pen stroke inside the patch.

4.3 Data augmentation

Paper has no preferred orientation, so every training and validation image is also used rotated by 90° and 180° and flipped horizontally, which quadruples the data. Test images are never augmented.

Chapter 5

Task (i): Three-Class Classification

5.1 Protocol

Every classifier is preceded by a `StandardScaler`. Hyper-parameters are chosen by grid search on the validation images (SVM: C and γ; Naive Bayes: variance smoothing; Logistic Regression: C; MLP: architecture and α). The selected model is refit on the augmented training and validation data and evaluated once on the 225 test images. Three feature sets were compared to measure the value of the candidate patch.

5.2 Results

Table 5.1: Test accuracy on 225 held-out images (validation accuracy in parentheses).

Classifier	Global (2,852)	Patch (2,107)	All (4,959)
SVM (RBF)	0.969 (0.973)	0.982 (**0.991**)	0.987 (0.987)
Naive Bayes	0.942 (0.920)	0.960 (0.960)	0.987 (0.964)
Logistic Regression	0.978 (0.973)	0.978 (0.973)	**0.991** (0.982)
Neural Network (MLP)	0.982 (0.969)	0.973 (0.982)	0.987 (0.978)

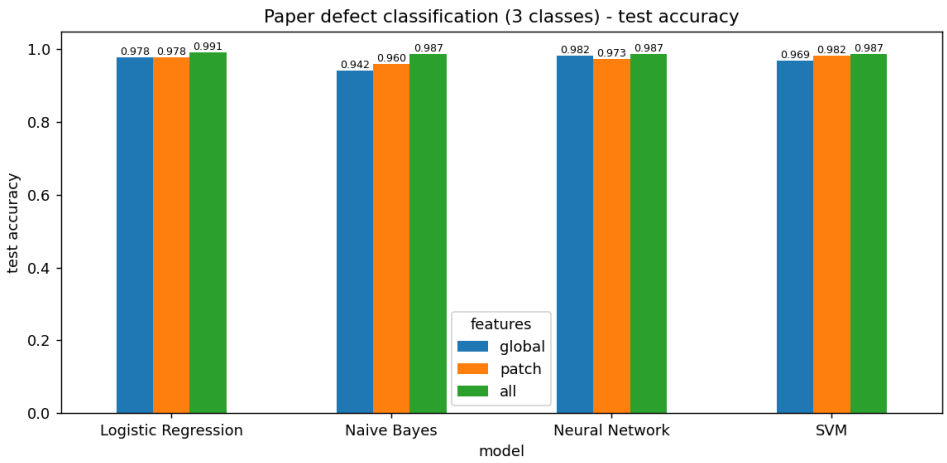


Figure 5.1: Test accuracy of the four classifiers for each feature set.

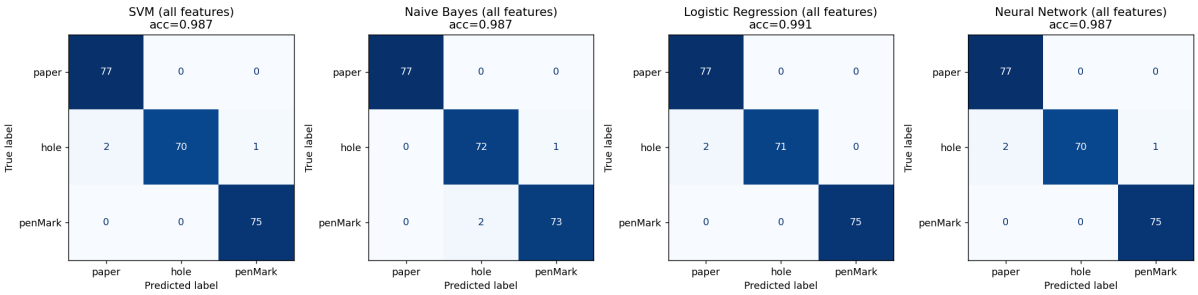


Figure 5.2: Test confusion matrices with all features.

Discussion. All four classifiers perform very well (94–99%) *because* the pre-processing makes the defects visible: on the illumination-flattened z-map, even the global HOG and spatial-binning features carry the defect signal. The candidate-patch features (texture and shape of the isolated object) give the highest validation accuracy, and combining both sets gives the best test accuracy (98.7–99.1%). With all features even Naive Bayes reaches 98.7%, which shows that the shape descriptors make the classes nearly linearly separable. The few remaining errors are almost all holes: very faint or very small holes that look like paper texture, and elongated holes that resemble a short pen stroke (Figure 6.2).

Chapter 6

Task (ii): Defect Localisation

6.1 Method

Localisation reuses the trained classifiers. A **region classifier** is trained on the patch features of the *top five* candidates of every training image:

- the top candidate of a hole or pen-mark image is labelled with the image class;
- all candidates of defect-free images, and the lower-ranked candidates of defect images (mostly static lens blemishes and texture specks), are labelled *background*. These act as **hard negatives** (5,235 training regions: 694 defects and 4,541 background).

At inference every candidate is scored, and the one with the highest $P(\text{defect})$ is the detected defect. Its bounding box, scaled back to 2000×2000 pixels, is drawn with the predicted class and confidence. The same decision also gives the image class (*multi-proposal decision*). A first version trained the region classifier only on top candidates. It was rejected after analysis because it labelled faint specks on clean paper as holes: it had never seen such regions during training. Hard-negative training fixed this.

Table 6.1: Region classifier (multi-proposal decision) accuracy.

Classifier	Validation	Test
SVM (RBF, $C = 10$, $\gamma = 10^{-4}$)	**0.996**	**0.973**
Naive Bayes	0.969	0.969
Logistic Regression ($C = 0.01$)	0.987	0.964
Neural Network (256, 64)	0.991	0.973

6.2 Final system

The multi-proposal SVM had the highest validation accuracy of all models (0.996), so it was selected as the final model for both tasks. An independent re-run of the saved model on the original test image files (`q3/verify_test.py`) reproduces the result:

Table 6.2: Final model on the 225 test images (accuracy 97.33%).

Class	Precision	Recall	F1	Images
Paper (no defect)	0.963	**1.000**	0.981	77
Hole	0.986	0.932	0.958	73
Pen mark	0.974	0.987	0.980	75
Overall	0.974	0.973	0.973	225

There are no false alarms on clean paper (77/77). 74 of 75 pen marks and 68 of 73 holes are recognised. The single-candidate classifiers of Table 5.1 are slightly better on test (up to 99.1%). The difference of four images is within the noise of a 225-image test set, and we report the validation-selected model to keep the protocol honest.

6.3 Localisation results

No ground-truth boxes are provided, so localisation was evaluated in two ways:

- **Consistency.** For 146 of the 148 defective test images, the chosen candidate is also the darkest region of the image. In the other two, the region classifier correctly passed over a static blemish.
- **Visual inspection** of zoomed crops (Figure 6.1) and of all misclassified images (Figure 6.2). In every inspected case the box lies tightly on the real hole or pen mark, including the six images whose class was wrong. Pen-mark boxes cover the whole stroke, because the morphological closing merges its fragments.

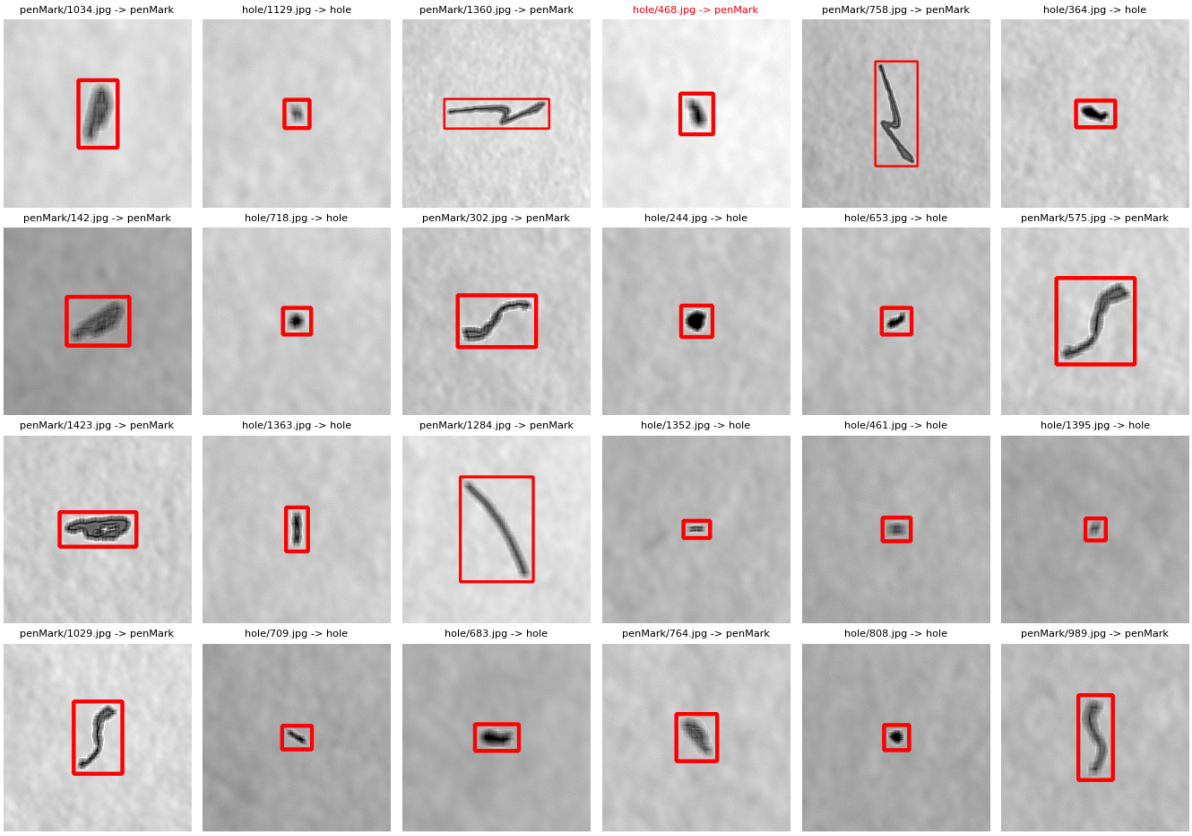


Figure 6.1: Detected bounding boxes on 24 random defective test images (zoomed; titles show the true class and the prediction).

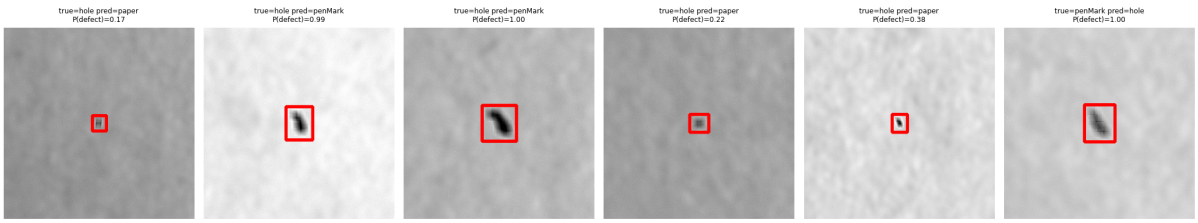


Figure 6.2: All misclassified test images. Each box is still on the defect; the errors are faint holes predicted as paper and elongated holes resembling pen strokes.

Chapter 7

Innovation: Interactive Inspection Application

The trained models are delivered as a web application (`streamlit run app.py`) that a quality-control operator could use directly.

- **Single-image inspection.** Upload a photo, or pick or randomly draw a test image. The app shows the class with its confidence, the probability of each class, the full image with the bounding box, the box coordinates in pixels and a zoom on the defect. For test images, the true class is shown for comparison.
- **Explanation: how the model sees the image.** An expandable panel shows the illumination-flattened z-map with all five candidate regions. For each candidate it lists the darkness score, the size and the region classifier's probabilities, and the chosen defect is highlighted. This makes it clear why a faint speck or lens blemish is rejected while the real defect is accepted.
- **Batch inspection.** Many images can be uploaded at once, as on a production line. The app reports the number of images, the number of defective ones and the defect rate, and the results table (class, confidence and box per image) can be downloaded as CSV.

Methodologically, the main innovation is the **proposal-based classical detector with hard negatives**. Instead of describing the whole image, the pipeline isolates candidate objects (illumination flattening, robust z-score, morphology) and learns to reject static lens blemishes, and a single classifier performs both classification and localisation.

Chapter 8

Software Design and Usage

```
ml_cep/
  app.py                        interactive inspection application
  q3/defect_features.py         pre-processing, candidates, all features
  q3/q3_defect_detection.py     split, augmentation, 4 classifiers x 3 feature sets,
                                region classifier, localisation, figures, models
  q3/predict.py                 command-line prediction + annotated images
  q3/verify_test.py             independent re-check of the final model on test images
  q3/error_analysis.py          figure of all misclassified test images
  q3/models/                    trained models
  q3/results/                   all tables and figures
  report/                       this report
```

```
pip install -r requirements.txt
# images: Google Drive folder from the handout -> q3/data/out/{paper,hole,penMark}/
python q3/q3_defect_detection.py      # full pipeline (~20 min on 12 cores)
python q3/verify_test.py              # final test accuracy from the saved model
python q3/predict.py image.jpg        # annotated image -> q3/predictions/
streamlit run app.py                  # application
```

Random seeds are fixed (42), and features are cached, so every number in this report can be regenerated. The trained models are included in the repository.

Chapter 9

Team Contribution

Table 9.1: Individual contributions.

Member	Main responsibilities
Sumayyah Mohammedy (SE-23013)	Literature review of surface-inspection features; dataset analysis and split; compilation and editing of the LaTeX report.
Amna Khan (SE-23039)	Pre-processing: illumination flattening, robust z-score map, candidate proposals and their calibration; data augmentation.
Mariha Mahmood (SE-23047)	Feature extraction (colour histogram, spatial binning, HOG, LBP, GLCM, shape and Hu moments); tuning and evaluation of the four classifiers on the three feature sets.
Haris Bin Nasir (SE-23084)	Region classifier with hard negatives and bounding-box localisation; error analysis and independent test verification; the Streamlit application and GitHub repository.

Table 9.2: Responsibility matrix (**L** = lead, **S** = support).

Task	Sumayyah	Amna	Mariha	Haris
Literature review, data analysis, split	L	S		
Pre-processing and candidates		L	S	
Features and classifiers		S	L	
Localisation and error analysis			S	L
Application and repository	S			L
Report	L	S	S	S

Chapter 10

Conclusion

We built a complete classical machine-vision pipeline for paper surface inspection. It fulfils all CEP requirements: the suggested features plus features from the literature, optional data augmentation, and four classifiers for two tasks. Flattening the illumination and isolating candidate regions turns a hard problem (defects below 0.01% of the image) into an easy one: every classifier exceeds 94% test accuracy and the best reaches **99.1%**. The final validation-selected system reaches **97.3%** with no false alarms, and localises the defect correctly in every inspected image.

Limitations and future work. The method assumes one dominant dark defect per image and dark defects on light paper. Bright defects such as back-lit tears would need a symmetric threshold. A quantitative IoU evaluation would require manually annotated boxes, and very faint holes remain the main error source; higher-resolution processing around candidates or a small CNN on the candidate patches could reduce these errors.

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Appendix A

Requirement Checklist

Table A.1: CEP requirements and where they are addressed.

Requirement	Where
Colour histogram, spatial binning, HOG	Ch. 4, Table 4.1
Literature review and further features	Ch. 4 (LBP, GLCM, shape, Hu moments)
Data augmentation (optional)	Section 4.3
SVM, Naive Bayes, Logistic Regression, Neural Network	Ch. 5, Table 5.1
(i) Three-class classification	Ch. 5, Table 6.2
(ii) Defect localisation with bounding boxes	Ch. 6, Fig. 6.1
Report, code, ease of use	this report, `https://github.com/hbnasir23/ml-cep-ct317`, Ch. 7

Appendix B

Hyper-parameter Grids

SVM (RBF)	C {0.3, 1, 3, 10, 30} x gamma {scale, 1e-4, 3e-4}
Naive Bayes	var_smoothing {1e-9, 1e-6, 1e-3, 1e-2, 1e-1}
Logistic Regression	C {0.001, 0.01, 0.1, 1}
Neural Network	hidden {(128), (256), (256, 64)} x alpha {1e-3, 1e-2, 1e-1, 1}